

VMAF-Aware Runtime Shielding for Constrained Deep Reinforcement Learning in Adaptive Video Streaming: A Strictly Pareto-Improving Projection

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Abstract

Client-side Adaptive Bitrate (ABR) control must consistently balance perceptual video quality against the risk of playback stalls. While purely reward-maximizing Deep Reinforcement Learning (DRL) can achieve strong average surrogate scores, it frequently learns aggressive bitrate policies that inflate tail stall risk under throughput volatility. Conversely, conventional runtime safety projections (“shields”) layered on top of such policies often guarantee stability only at the *unauditable* expense of visual quality, typically by forcing opaque uniform index decrements that are hard for operators to retune.

We formulate DASH/HLS bitrate selection as a Constrained Markov Decision Process (CMDP) that maximizes expected perceptual fidelity (VMAF in our harness) subject to empirical budgets on rebuffer accumulation and smoothed quality switching, instantiated with primal–dual Lagrangian relaxation and Proximal Policy Optimization (PPO) together with manifest lookahead and buffer-stabilization cues. On top of the learned policy we introduce a *VMAF-aware runtime shield*: a deterministic action-projection layer that, whenever emergency step-down is required, navigates the bitrate ladder using *non-linear, content-dependent* VMAF spacing (with an explicit VMAF-loss budget and a tunable catastrophic-ratio knob) rather than naive index-only decrements.

Extensive trace-driven replay on a standard FCC/Norway-style parti-

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tioned suite ($n=80$ paired episodes on identical policy and seeds) shows that the proposed architecture *Pareto-improves* legacy index-decrement shielding in mean QoE, rebuffering, and VMAF, with a strictly significant triple at tighter VMAF-loss tolerance $\tau=0.8$ and a marginal VMAF p -value at the recommended $\tau=1.0$. For example, at VMAF-loss tolerance $\tau=0.8$, the VMAF-aware shield suppresses mean session rebuffering from 11.04 % without the shield to 4.98 %, compared with 5.67 % for legacy projection, while simultaneously raising mean VMAF (66.13 vs. 65.82) and mean session QoE (2583.70 vs. 2518.87). At the recommended $\tau=1.0$ operating point, paired Wilcoxon tests confirm large QoE and rebuffer gains versus legacy ($p=5.0 \times 10^{-3}$, $p=2.9 \times 10^{-3}$) with a mean VMAF improvement whose two-sided p -value is marginal ($p=0.056$; $n=80$). Tightening to $\tau=0.8$ yields a strictly significant triple (+64.82 Δ QoE, -0.69 % rebuffer, $++0.32$ VMAF; $p=1.4 \times 10^{-3}$, $p=2.7 \times 10^{-3}$, $p=4.1 \times 10^{-2}$). Across 47 shield configurations the sweep empirically collapses to only 11 behavioral classes, isolating the VMAF-aware fallback as the dominant Pareto driver.

We further stress-test the same policy on an *out-of-distribution* suite of synthetic **5G/mmWave-oriented stress traces** with intermittent outages: mean rebuffering grows to ≈ 70.26 % without protection, while VMAF-aware projection preserves large paired QoE lifts and statistically significant rebuffer reductions relative to legacy projection (e.g. $+34.60$ – $+39.85$ Δ QoE; -2.27 %– -2.38 percentage-point Δ rebuffer; $n=80$); paired mean VMAF is not cleanly resolved there, as expected when sessions become stall-saturated.

Removing the shield at inference from a shield-co-trained policy inflates rebuffering by $|-5.92|$ % ($p=1.4 \times 10^{-10}$), underscoring policy–shield co-design. Relative to Pensieve and RobustMPC under the same harness, the recommended operating point reduces mean rebuffering by 9.87 % and 25.58 %, respectively. Overall, the VMAF-aware projection is a quality-preserving, statistically grounded safety layer for constrained DRL ABR, validated on broadband partitions and corroborated under mobile-oriented outage stress.

Keywords: Adaptive Bitrate (ABR) Streaming, Deep Reinforcement Learning (DRL), Constrained Markov Decision Process (CMDP), Runtime Shielding, VMAF-aware optimization, 5G networks, Lagrangian duality, Proximal Policy Optimization, Policy–shield co-design, Video QoE

1. Introduction

Deep client-side *adaptive bitrate* (ABR) policies trained with reinforcement learning can track high average surrogate rewards, yet their **greedy tail behavior** is precisely what mobile deployments fear [1, 2]. Sub-6 GHz and especially **5G mmWave** access routinely inject millisecond-to-second capacity cliffs from blockage, LOS/NLOS transitions, scheduling, and mobility [3, 4]. Under such volatility, unconstrained Deep RL (DRL) often remains *bitrate-hungry* long enough to convert rare throughput dips into **long tail stalls** that dominate viewer Quality of Experience (QoE) even when mean logs look acceptable [5, 6]. **The opening crisis is operational, not academic:** aggressive learned controllers can “win” average scores while silently concentrating probability mass on catastrophic rebuffer events [1].

Industry-grade stacks therefore bolt on *runtime shields*—deterministic action projections executed immediately before each chunk download [7]. **The dominant legacy design is *content blind*.** When buffer-depletion risk triggers, conventional shields—including the index-decrement projection we inherit as a V12-era baseline—almost always apply **representation-index decrements**: step down one ladder rung, repeat until a crude viability test passes [1, 7]. Because perceptual fidelity is **not** affine in discrete bitrate indices, those decrements routinely *over-pay* in VMAF for modest safety gains: adjacent ladder steps can differ by a few VMAF points or by ten, depending on content and encoding [8]. **Legacy shielding is therefore a stopgap:** it enforces a crude safety-quality *trade-off*—operators accept opaque visual damage to avoid stalls—even when a safer downgrade path with negligible perceptual loss exists on the same manifest [9, 10].

We change the paradigm: the shield is itself an optimization layer, not merely a panic brake. Concretely, we introduce a **VMAF-aware runtime shield**: a deterministic projection rule that, whenever a catastrophic-download predicate fires, searches downward along the *manifest-known* bitrate ladder under explicit buffer-margin viability constraints while maximizing retained VMAF subject to a perceptual-loss budget τ and a tunable catastrophic-ratio knob ρ_{cat} (Sec. 4.4). The logic is **analytically simple yet perceptual**: it is a transparent, auditable function of chunk sizes, conservative throughput surrogates, buffer seconds, and per-chunk VMAF tables—not another opaque neural head [7, 8]. **Our method does not force operators to forfeit perceptual fidelity for safety by default;** it replaces blind index hopping with ladder-respecting downgrade choices that

preserve as much VMAF as emergencies allow.

Headline evidence: paired Pareto gains with rigorous statistics, not another hand-tuned compromise. Holding the trained policy, videos, and trace seeds fixed, we pair $n=80$ replay episodes and test legacy projection against VMAF-aware projection. **On the standard FCC/Norway-style partitioned throughput suite** central to trace-driven ABR research, Wilcoxon signed-rank tests show **simultaneous mean improvements** in QoE, rebuffering, and VMAF versus legacy on paired episodes, with QoE and rebuffering highly significant at $\tau=1.0$ ($p=5.0\times 10^{-3}$, $p=2.9\times 10^{-3}$) and VMAF marginal ($p=0.056$; Sec. 5). Tightening τ from 1.0 to 0.8 yields a **strict triple** significant at conventional $\alpha=0.05$ (+64.82 QoE, -0.69% rebuffer, $++0.32$ VMAF; $p=1.4\times 10^{-3}$, 2.7×10^{-3} , 4.1×10^{-2}). **We then stress the identical policy on synthetic 5G/mmWave-style outage traces** ($n=80$ paired replay) to mimic blockage-driven volatility [3]: unshielded operation collapses into massive rebuffering, whereas VMAF-aware projection retains **large paired QoE gains and highly significant rebuffer reductions versus legacy** (Sec. 5.9). There, session-mean VMAF tracks legacy within fractions of a point and two-sided Wilcoxon tests do not separate paired VMAF at conventional levels—an expected artifact when playback becomes stall-saturated rather than a contradiction of the primary strict-Pareto certificate [11]. Finally, shield-design sweeps collapse 47 configurations into only 11 behavioral classes dominated by the VMAF-aware fallback, and removing the shield at inference from a shield-co-trained policy inflates rebuffering by $|-5.92|\%$ ($p=1.4\times 10^{-10}$), proving empirical **policy–shield co-design** [7].

This paper contributes the following:

- **CMDP formulation and Lagrangian PPO training.** We cast ABR as a Constrained Markov Decision Process with primal–dual updates over stall and switching surrogates, Proximal Policy Optimization (PPO), manifest lookahead, and buffer-regularization cues [12–16].
- **VMAF-aware runtime shielding.** We design a deterministic projection layer that optimizes emergency downgrade paths along nonlinear perceptual ladders instead of naive index decrements, with auditable predicates and operator-facing knobs τ and ρ_{cat} (Sec. 4.4).
- **Paired statistical evidence.** Wilcoxon tests on $n=80$ episodes establish significant QoE and rebuffer gains vs. legacy at $\tau=1.0$, a strictly

significant triple at $\tau=0.8$, and synthetic 5G/mmWave stress replay corroborates large QoE and stall reductions (Sec. 5–5.9).

- **Inference co-design evidence.** We show empirically that constrained policies learn bitrate proposals whose safety is conditional on the deterministic shield at deployment time ($| - 5.92 |$ % paired stall inflation when the shield is removed; $p=1.4 \times 10^{-10}$) [1, 7].

Road map. Section 2 reviews related ABR and safe RL literature; Section 3 formalizes the QoE surrogate, CMDP, and shield interfaces; Section 4 details Lagrangian PPO and VMAF-aware projection; Section 5 presents paired statistics, tables, and figures (primary suite plus 5G/mmWave stress); Sections 6–7 discuss limitations and conclusions.

2. Related Work

We structure the literature into three streams that culminate in the research gap our VMAF-aware shield targets. **Scope.** Our contribution centers on *client-side bitrate / action selection* under manifest and buffer observables—the control policy and its deterministic runtime projection—not on inventing a new state-of-the-art throughput estimator: data-driven predictors such as CS2P [17] remain complementary infrastructure that any controller (including ours) may consume [1, 18]. Recent surveys [5, 19] and 5G/mmWave streaming measurement studies [3] motivate why volatility-aware *control* matters independently of marginal gains in mean throughput forecasts.

2.1. Heuristic and model-based ABR (interpretable, tuning-limited)

MPEG-DASH decouples packaging from transport; clients run local policies from ladder metadata and download observations [15]. Buffer-threshold heuristics such as BBA [20] achieve remarkable simplicity and transparency, while BOLA-style online formulations expose Lyapunov-structured stability arguments for buffer evolution [21]. RobustMPC [18] epitomizes predictor-assisted constrained lookahead: bitrate is chosen by solving a short-horizon program under an estimated throughput trace. **Strengths.** These designs are comparatively auditable: operators can reason about thresholds, horizons, and constraint violations without opening a neural graph. **Limits toward 5G-class volatility.** Hand tuning does not scale across CDN regions,

device classes, and blockage-heavy mmWave dynamics [3, 4]; when predictors lag abrupt capacity cliffs, even principled MPC objectives can mis-fire, and purely heuristic guards rarely exploit nonlinear perceptual ladders [8]. Recent SIGCOMM work on smoothness-aware online control (SODA [22]) sharpens production-relevant consistency objectives yet remains outside the DRL+shield co-design space we target. Orthogonal systems layers—edge assistance [23–25] and Oboe-style ABR auto-tuning [26]—change *where* or *how* knobs are calibrated, but do not by themselves resolve greedy neural tails or content-blind emergency projection.

2.2. Deep reinforcement learning for ABR (*expressive, opaque, tail-risk prone*)

Pensieve [1] catalyzed trace-driven deep RL for ABR; contemporaneous and follow-on DRL formulations (e.g., D-DASH [27]) demonstrate that neural policies can absorb heterogeneous histories and manifest features [2]. Recent PPO-flavored streaming RL [28] and hybrid predictor–learner pipelines [29] continue this trend into the 2023–2024 window emphasized by networking venues [5]. **Strengths.** Scalarized RL objectives can track high average QoE surrogates on curated traces. **Critical tension.** The policy network is a *black box* to operators: maximizing expected return encourages aggressive bitrate choices whose failures concentrate in rare but catastrophic rebuffer tails under bursty cellular links [1, 6]. Field deployments (Fugu / learning in situ [6]) further highlight distribution shift between simulators and live access networks. **Hand-off to constrained learning.** Declarative stall and smoothness budgets—rather than implicit reward scaling—motivate CMDP and safe RL viewpoints for streaming [9, 10].

2.3. Safe / constrained RL, shields, and the content-blindness gap

CMDPs and Lagrangian / primal–dual RL [12, 13, 30] supply the optimization vocabulary for constraint-aware training; tutorial taxonomies categorize shields, Lyapunov methods, and risk-aware policy search [10, 31]. QoE-oriented safe RL studies increasingly import these ideas into streaming [9]. Runtime shielding [7] repairs or projects actions before environmental execution, yielding measurable intervention telemetry. **The unresolved gap.** In practice, streaming shields (including legacy index-decrement projection baselines in our own experimental lineage) overwhelmingly implement **content-blind** *representation-index decrements*: when danger is detected, the client steps down the discrete ladder without asking how nonlinear VMAF spacing couples that decrement to perceptual damage [8]. Such

Table 1: High-level positioning for *client-side control*: deterministic pre-fetch projection (“Det. guard”) and whether emergency logic is **VMAF-ladder-aware** (not merely bitrate-index-aware). $\approx$ indicates constraint-aware training without a focal logged shield.

Representative work	Year	Paradigm	Det. guard	VMAF-aware
BBA [20]	2014	Heuristic	N	N
RobustMPC [18]	2015	MPC / lookahead	N	N
Pensieve [1]	2017	DRL	N	N
Oboe [26]	2018	Meta-tuning	N	N
Fugu / in-situ [6]	2020	Field DRL	N	N
SODA [22]	2024	Online control	N	N
QoE safe RL [9]	2025	Constr. DRL	$\approx$	N
Ours (Lag. PPO + VMAF shield)	—	CMDP + shield	Y	Y

Det. guard: deterministic action projection immediately before chunk download with auditable predicates (shield), distinct from stochastic policy sampling alone [7]. **VMAF-aware:** emergency search uses per-chunk VMAF ladder spacing, not only uniform index decrements [8].

shields behave like emergency brakes—they trade away opaque amounts of visual fidelity to buy safety—even when an alternative feasible downgrade path could preserve nearly all VMAF [9, 10]. **Our departure.** We treat the fallback itself as a perceptual optimization primitive: a deterministic VMAF-aware projection layer atop Lagrangian PPO, rather than another implicit QoE–stall compromise baked into reward weights alone.

Punchline. In contrast to literature that treats safety projection primarily as a content-blind panic mechanism, we shift the paradigm: the shield *actively optimizes* the emergency downgrade trajectory subject to buffer-feasibility tests, using manifest-known VMAF ladders [7, 8]. To our knowledge, ours is the first trace-driven ABR architecture in our benchmarking harness to embed nonlinear perceptual metrics inside a deterministic runtime shield and to demonstrate *strict empirical Pareto improvement* (paired QoE, rebuffer, VMAF) over legacy projection on identical policies and seeds (Sec. 5).

Table 1 summarizes the gap; Table 2 gives a finer-grained feature taxonomy (stall budgets, lookahead, neural map). Edge-assisted delivery [24, 25] and predictor innovations [17] stack orthogonally atop the control layer we focus on.

Table 2: Finer-grained taxonomy of representative client-side ABR paradigms (complements the positioning grid of Table 1; symbol legend below).

Family (ref.)	Summarized objective / setting	Stall/ smooth budget	Manifest lookahead	Neural bitrate map	Runtime determ. guard
BBA [20]	Thresholds on buffer occupancy	Ind.	Imp. sizes	N	Heuristic thresh- olds
BOLA [21]	Online Lyapunov-structured bitrate utility	Imp. stabil- ity lens	Implicit (buffer cou- pling)	N	N
RobustMPC [18]	Short-horizon MPC under predicted throughput	Ind.	Horizon chunk sizes	N	N
Oboe [26]	System-level ABR parameter auto-tuning	N	N	N	N
Pensieve [1]	Scalar RL reward (QoE surrogate)	NF	Var. (blind- ing op- tional)	Y	N
D-DASH [27]	DRL on DASH traces	NF	Manifest ladder fea- tures	Y	N
Fugu / in-situ [6]	Field RL on live sessions	Imp.	Standard client ob- serv- ables	Y	N
QoE-aware safe RL [9]	Constrained / safe RL for streaming QoE	Focal	State design depend- ent	Y	Ctx
Hybrid control- learning [29]	Structured predictors plus learning	Var.	Often predictor- driven	P	N
Ours (Lag. PPO + optional shield)	Primal-dual CMDP surrogate trained with PPO [13, 14]	Adapt. dual λ	MLP policy + mani- fest LA [1, 15]	Y	Shield (opt.) [7]

Column symbols. **Y**=feature clearly present *in spirit*; **N**=not focal; **Ind.**=indirect via rewards/structure; **Imp.**=implicit through dynamics/buffer coupling; **NF**=no declarative constrained stall/smooth budget like ours; **Var.**=configuration-dependent; **P**=partial hybrid coupling; **Ctx**=context-dependent guarding. BOLA: analytic Lyapunov stabilization [21] (not neural CMDP).

3. System Model and Problem Formulation

3.1. Streaming architecture and control loop

We adopt the standard HTTP adaptive-streaming abstraction [1, 15, 18]: a *video origin/CDN* exposes a discrete bitrate ladder $\mathcal{B} = \{0, \dots, L-1\}$ with per-chunk sizes and perceptual metadata in the DASH/HLS manifest; the *client* maintains a playout buffer B_k and, at each chunk boundary k , requests the next segment over a time-varying bottleneck whose effective throughput $\hat{C}_k$ is exogenous to the controller in trace-driven emulation. **Mobile volatility.** On sub-6 GHz and mmWave access, $\hat{C}_k$ can jump within milliseconds under blockage, LOS/NLOS transitions, and scheduling [3, 4]; this is the regime where greedy learned policies concentrate probability on *tail stalls* despite benign average traces [1, 6].

Decision pipeline. A neural policy π_θ maps an augmented state s_k to a stochastic *proposed* discrete action $a_k^{\text{raw}} \in \mathcal{B}$ (sampled or argmax during evaluation). Before the download of chunk $k+1$ begins, a **VMAF-aware runtime shield** $\mathcal{S}$ applies a deterministic map

$$(a_k^{\text{exec}}, \iota_k) = \mathcal{S}(s_k, a_k^{\text{raw}}), \quad a_k^{\text{exec}} \in \mathcal{B}, \quad \iota_k \in \{0, 1\}, \quad (1)$$

yielding the *executed* representation index a_k^{exec} and an audit bit ι_k indicating whether projection altered the proposal. Only a_k^{exec} is passed to the chunk download and buffer dynamics (4); Figure 4.2 highlights this $a_k^{\text{raw}} \rightarrow \mathcal{S} \rightarrow a_k^{\text{exec}}$ path. Decisions evolve on the chunk clock (playback duration T_c per step, e.g. 4 s in our harness) [1, 18].

Soft vs. hard safety (preview). Lagrangian PPO (below) encodes stall/smoothness *budgets* as **soft** expectations learned over many rollouts: dual variables λ_r, λ_s reweight penalties but a neural policy alone offers no per-step certificate that a dangerously large representation will never be attempted—especially under out-of-training 5G-style swings [3, 13, 30]. The shield supplies a **hard, deterministic** repair map at every step: whatever π_θ proposes, the client executes a manifest-feasible a_k^{exec} that respects explicit buffer-margin viability tests before committing bandwidth [7].

3.2. State, action, reward, and nonlinear perceptual mapping

State s_k . Beyond classical telemetry—buffer occupancy, past throughput samples, and the previous decision—our client observation includes **manifest lookahead**: near-future chunk sizes across the ladder and spatial/temporal

complexity tags (SI/TI) aligned with MPC-style horizon inputs [1, 15, 18]. Crucially for VMAF-aware shielding, the state interface also exposes (or implies through manifest tables) the **per-chunk VMAF ladder** $\{V_k(j)\}_{j \in \mathcal{B}}$: for chunk k , $V_k(j)$ is the objective perceptual score if representation j is fetched [8]. If either the policy or the shield ignored $\{V_k(j)\}$, content-dependent ladder spacing could not inform emergency downgrade, contradicting the premise of this paper.

Nonlinear, content-dependent fidelity. For each chunk k , the map $j \mapsto V_k(j)$ is **not** an affine function of the discrete index j , nor is it time-invariant across k : different GOP/texture structure yields different marginal VMAF gains between adjacent rungs. Equivalently, the composition “nominal bitrate $\rightarrow$ VMAF” is a **nonlinear, piecewise, content-varying** mapping; this is the mathematical reason legacy *index-decrement* shields can inflict disproportionate perceptual loss relative to a VMAF-aware search over the same feasible set [8].

Action. $a_k^{\text{raw}} \in \mathcal{B}$ is the policy’s proposed bitrate index for chunk $k+1$; the environment observes only a_k^{exec} .

Benchmarking reward (QoE surrogate). Following RobustMPC/Pensieve reporting conventions [1, 18], we log a scalar per-chunk surrogate with fidelity $q_k \equiv V_k(a_k^{\text{exec}})$:

$$\text{QoE}_k^{(\text{rep})} = q_k - \alpha \text{smooth}_k - \beta \tau_k^{\text{stall}}, \quad (2)$$

where $\text{smooth}_k = |q_k - q_{k-1}|$ proxies quality switching, τ_k^{stall} is rebuffer seconds on step k , and α, β match the shared evaluation harness. Session QoE is $\sum_{k \in \mathcal{K}_e} \text{QoE}_k^{(\text{rep})}$ as in Section 5. The *CMDP objective* below separates primal fidelity $\sum q_k$ from explicit constraint channels on stall/smooth statistics [12, 13].

3.3. Chunk-timescale dynamics

3.3.1. Throughput and latency

Let $\hat{C}_k > 0$ denote the effective throughput for chunk $k+1$ (trace-driven, comparable to baselines). Choosing a_k^{exec} induces payload size $\sigma_k(a_k^{\text{exec}})$ from the manifest; with $\tilde{C}_k = \max\{\hat{C}_k, \epsilon\}$, $\epsilon > 0$, nominal download seconds are $d_k = \sigma_k(a_k^{\text{exec}})/\tilde{C}_k$ [18].

3.3.2. Buffer and stall

Let $B_k \geq 0$ be buffer seconds at the start of download k [1, 18]. Rebuffering (stall) and buffer update follow

$$\tau_k^{\text{stall}} = \max\{0, d_k - B_k\}, \quad (3)$$

$$B_{k+1} = \max\{0, B_k - d_k\} + T_c. \quad (4)$$

Stochasticity enters through $\hat{C}_k$ on curated traces [19, 32, 33]; given $(s_k, a_k^{\text{exec}}, \hat{C}_k)$, transitions are deterministic.

3.4. CMDP costs, safety budgets, and Lagrangian relaxation

From MDP to CMDP. An unconstrained MDP maximizes $\mathbb{E}[\sum_k \gamma^k r_k]$ alone. We treat streaming as a **Constrained MDP**: maximize expected fidelity subject to budgets on *costs* induced by the same trajectory [12]. Define instantaneous stall cost $c_k^{\text{stall}} \equiv \tau_k^{\text{stall}}$ (seconds of stall on step k) and smoothness cost $c_k^{\text{sm}} \equiv \text{smooth}_k$. A canonical discounted formulation requires, for discount $\gamma \in (0, 1]$ and budgets $d_{\text{rf}}, d_{\text{sm}}$,

$$\mathbb{E}_{\pi_{\theta}, \rho} \left[\sum_{k \in \mathcal{K}_e} \gamma^{k-1} c_k^{\text{stall}} \right] \leq d_{\text{rf}}, \quad \mathbb{E}_{\pi_{\theta}, \rho} \left[\sum_{k \in \mathcal{K}_e} \gamma^{k-1} c_k^{\text{sm}} \right] \leq d_{\text{sm}}, \quad (5)$$

Artifact alignment. Our implementation follows the Pensieve-lineage *undiscounted episodic* summaries used for logging and dual updates: normalized stall ratio $\text{RB}(e)$ and average $\text{SM}(e)$, which are positive linear functionals of $\{c_k^{\text{stall}}\}$ and $\{c_k^{\text{sm}}\}$ over $\mathcal{K}_e = \{1, \dots, K_e\}$ [1, 13]. Let $T_e^{\text{play}} = K_e T_c$. Then

$$\text{RB}(e) = \frac{1}{T_e^{\text{play}}} \sum_{k \in \mathcal{K}_e} c_k^{\text{stall}}, \quad \text{SM}(e) = \frac{1}{K_e} \sum_{k \in \mathcal{K}_e} c_k^{\text{sm}}. \quad (6)$$

The constrained program motivating training is

$$\max_{\pi_{\theta}} \mathbb{E}_{\pi_{\theta}, \rho} \left[\sum_{k \in \mathcal{K}_e} q_k \right], \quad (7)$$

$$\text{s.t. } \mathbb{E}_{\pi_{\theta}, \rho} [\text{RB}(e)] \leq \delta_{\text{rf}}, \quad (8)$$

$$\mathbb{E}_{\pi_{\theta}, \rho} [\text{SM}(e)] \leq \delta_{\text{sm}}, \quad (9)$$

with targets $\delta_{\text{rf}}, \delta_{\text{sm}}$ fixed for reproducibility [18].

Lagrangian (soft) relaxation. Because directly enforcing (8)–(9) inside non-convex policy search is difficult, we form the Lagrangian in per-step penalty channels $\widetilde{\text{RB}}_k, \widetilde{\text{SM}}_k$ matching the codebase rollouts:

$$\mathcal{L}(\theta, \lambda) = \mathbb{E}_{\pi_{\theta}, \rho} \left[\sum_{k \in \mathcal{K}_e} (q_k - \lambda_r \widetilde{\text{RB}}_k - \lambda_s \widetilde{\text{SM}}_k) \right], \quad (10)$$

with $\lambda_r, \lambda_s \geq 0$ adapted by projected dual ascent on empirical gaps $g_r = \widehat{\text{RB}} - \delta_{\text{rf}}, g_s = \widehat{\text{SM}} - \delta_{\text{sm}}$ [12, 13]. Equivalently, the *instantaneous* reward used inside PPO can be read as $q_k - \lambda_r \widetilde{\text{RB}}_k - \lambda_s \widetilde{\text{SM}}_k$ (plus optional stabilizers; Sec. 4)—a standard “reward minus $\lambda \cdot \text{cost}$ ” construction. Finite-sample PPO with clipped objectives does not certify constraint satisfaction at every iterate [14, 30]; that gap motivates the hard shield in (1).

Remark (auxiliary Lyapunov cues). Training optionally subtracts buffer-deviation and simulator Lyapunov penalties [16] outside (λ_r, λ_s) ; they are stabilizers, not formal certificates.

3.5. VMAF-aware shield as a hard projection

Let $\mathcal{A}_{\text{safe},k}(a_k^{\text{raw}}) \subseteq \mathcal{B}$ denote the set of representation indices that pass the shield’s buffer-margin and catastrophic-download viability tests at step k (Sec. 4.4), instantiated with pessimistic throughput, manifest sizes $\sigma_k(\cdot)$, and buffer B_k . **Legacy content-blind repair** maps a_k^{raw} downward by repeated index decrements until a feasible index is found [7].

VMAF-aware projection (conceptual). Among feasible rungs we retain maximal perceptual score subject to a small VMAF-loss budget τ relative to the proposal:

$$a_k^{\text{exec}} \in \arg \max_{j \in \mathcal{A}_{\text{safe},k}(a_k^{\text{raw}}) : V_k(a_k^{\text{raw}}) - V_k(j) \leq \tau} V_k(j), \quad (11)$$

with a strict light/strong fallback if (11) is empty [8]. Thus $\mathcal{S}$ is precisely a **deterministic perceptual projection** $\pi_{\text{shield}} : (s_k, a_k^{\text{raw}}) \mapsto a_k^{\text{exec}}$ optimizing retained VMAF on $\mathcal{A}_{\text{safe},k}$ instead of blindly stepping down the index order. Full predicate details (catastrophic ratio ρ_{cat} , pessimism ρ , margin μ) appear in Section 4.4.

4. Method: Lagrangian PPO and Runtime Shields

This section specifies an implementation-faithful recipe so third parties can *reproduce* training and shielding: Stable-Baselines 3 PPO [14, 34] with

our Lagrangian `gymnasium.Wrapper`, plus deterministic projection `safe_adjust_action_v13` in `safety_shield_v13.py` (primary path for `train_all_models_v13.py` / `evaluate_all_models_v13.py`), and—for VMAF-aware online sweeps—`safe_adjust_action` in `safety_shield_v12.py` with `vmaf_aware=True`. Symbols match Sec. 3: horizon $k = 1, \dots, K_e$, T_c , $a_k^{\text{raw}}, a_k^{\text{exec}} \in \mathcal{B}$, B_k , τ_k^{stall} , buffering (4), and logged penalties $\widetilde{\text{RB}}_k, \widetilde{\text{SM}}_k$.

4.1. Observation interface and policy outputs

State s_k and the shield map $\mathcal{S}$ are defined in Section 3; the actor-critic stack consumes the same tensors as Figure 4.2 (throughput history, buffer/last action, manifest lookahead including SI/TI and per-chunk VMAF ladders) [1, 8, 15, 18]. The actor outputs a categorical distribution over raw proposals $a_k^{\text{raw}} \in \mathcal{B}$; optional hysteresis or shield-aware shaping wrappers may alter sampling incentives, but the **hard** executed action is always $a_k^{\text{exec}} = \mathcal{S}(s_k, a_k^{\text{raw}})$ when shielding is enabled [7].

4.2. Lagrangian PPO agent with Lyapunov-style buffer regularization

Baseline optimizer (standard PPO). We deliberately *do not* claim algorithmic novelty for clipped PPO itself [14]: twin Tanh MLP heads parameterize $\pi_\theta(a \mid s_k)$ and $V_\phi(s_k)$; vectorized rollouts use GAE ($\gamma, \lambda_{\text{GAE}}$), surrogate clipping, entropy bonuses, and Adam schedules frozen in Table 3 [34].

Where this paper invests design effort: CMDP shaping + stability cues. The train-time signal is the Lagrangian-relaxed objective (10) instantiated per step by (12). Besides dual-weighted stall/smooth channels, we inject **Lyapunov-style buffer regularizers** $\phi(B_k)$ and emulator Lyapunov telemetry ψ_k^{Lya} with fixed coefficients (w_b, w_ℓ) *outside* the dual pair (λ_r, λ_s) [16, 31]. Intuitively, ϕ penalizes dangerous buffer deviations early in training, tempering exploratory bitrate spikes before dual variables fully adapt; ψ^{Lya} surfaces simulator-provided drift cues analogous to queue-stability heuristics. These terms are **not** formal certificates for neural policies; they reduce variance and premature dual blow-ups in trace-driven CMDP+PPO training [10, 13].

End-to-end control loop (cf. Sec. 3): the actor proposes a_k^{raw} ; the VMAF-aware shield $\mathcal{S}$ projects it to a_k^{exec} before the chunk download updates the client buffer under time-varying throughput. Manifest lookahead (including ladder VMAF tables) feeds the augmented state s_k ; auxiliary penalties (ϕ, ψ^{Lya}) enter Lagrangian training only.

4.2.1. Lagrangian reward channel and dual ascent

Each step logs $(q_k, \text{rebuffer}_k, \text{smooth}_k, \phi(B_k), \psi_k^{\text{Lya}})$ into **info**; $\widetilde{\text{RB}}_k, \widetilde{\text{SM}}_k$ match the stall/smooth penalties in (10). The PPO surrogate uses (repository scaling 1/100 for SB3 numerics)

$$r_k^{\text{train}} = \frac{1}{100} \left(q_k - \lambda_r \widetilde{\text{RB}}_k - \lambda_s \widetilde{\text{SM}}_k - w_b \phi(B_k) - w_\ell \psi_k^{\text{Lya}} \right). \quad (12)$$

After warm-up ($H=60$ episodes in the V12 freeze), episodic $\hat{\text{RB}}, \hat{\text{SM}}$ drive asymmetric projected dual ascent on gaps $g_r = \hat{\text{RB}} - \delta_{\text{rf}}, g_s = \hat{\text{SM}} - \delta_{\text{sm}}$ with caps $[\lambda^{\min}, \lambda^{\max}]$ per axis [10, 13]; vectorized workers share or replicate $\lambda_\bullet$ per artifact defaults.

4.3. Catastrophic risk detection and pessimistic feasibility

The shields never learn parameters; each evaluates **deterministic** predicates on manifest data and a conservative kbps estimate. Let $\overline{\text{tp}}_k$ be the instantaneous trace throughput at chunk k and $\underline{C}_k$ the *pessimistic* kbps estimate

$$\underline{C}_k = \max \left\{ C_{\min}, \rho \cdot \min \{ \overline{\text{tp}}_k, \text{last_tp}_k \} \right\}, \quad (13)$$

with pessimism $\rho \in (0, 1]$ and emulator throughput floor $C_{\min}$ matching the codebase (**V13** defaults $\rho=0.97$; frozen **V12** VMAF-sweep configs often use $\rho=0.90$). For candidate representation j , predicted download seconds are

$$d_k^{\text{est}}(j) = \min \left\{ \frac{\sigma_k(j)}{\underline{C}_k \cdot 10^3}, 60 \right\}, \quad (14)$$

where $\sigma_k(j)$ is manifest chunk size in bits for bitrate index j at chunk k . A fluid analogue of post-download buffer (ignoring simultaneous playback credit during download, but mirroring conservative tests) is

$$\widetilde{B}_{k+1}(j) = \max \{ 0, B_k - d_k^{\text{est}}(j) \} + T_c, \quad (15)$$

which operators can compare against a **safety floor** B_{safe} : if $\widetilde{B}_{k+1}(j)$ is routinely below B_{safe} , representation j is emergency-infeasible. **Risk gate**. When enabled (**only_when_risky=True** in V13), the shield *passes through* a_k^{raw} whenever

$$d_k^{\text{est}}(a_k^{\text{raw}}) \leq \max \{ B_k, 0.1 \} \cdot \rho_{\text{risk}}, \quad (16)$$

with $\rho_{\text{risk}}=1.35$ in the V13 freeze (versus 1.10 in many V12 configurations); benign proposals therefore skip projection [7]. **Catastrophic pass-through**.

In V12 with `vmaf_aware=True`, if $d_k^{\text{est}}(a_k^{\text{raw}}) \leq B_k \rho_{\text{cat}}$ the implementation returns $a_k^{\text{exec}} = a_k^{\text{raw}}$ without constructing the VMAF pool (Algorithm 2). The V13 **light** path uses the same inequality as a cheap no-intervention check before margin-based repair. **Critical buffer.** If $B_k \leq B_{\text{crit}}$, the shield snaps to index 0: **V13** uses $B_{\text{crit}}=0.20$ s, whereas **V12** uses 0.30 s—both are hard emergency rules.

4.4. Deterministic runtime projection

Role. $\mathcal{S} : (s_k, a_k^{\text{raw}}) \mapsto (a_k^{\text{exec}}, \iota_k)$ executes *before* `env.step`; $\iota_k \in \{0, 1\}$ logs interventions for audit CSVs (Section 5).

4.4.1. QoE-oriented shield core (V13)

Besides **off**, V13 exposes **qoe**, **light**, and **strong**. All modes share $\underline{C}_k$ and $d_k^{\text{est}}(\cdot)$ from (13)–(14), clip a_k^{raw} to $\mathcal{B}$, apply the critical-buffer snap, and honor the optional risk gate (16). The **qoe** mode searches downward from a_k^{raw} over at most Δ_{dn} index steps (default 2), but *accepts* representations whose predicted download fits a small **predicted-stall budget** ζ (default 0.25 s), i.e. $d_k^{\text{est}}(j) \leq B_k + \zeta$, before widening search down to a minimum guard index j_{min} (default 1). If no candidate satisfies the budget, V13 may emit an emergency index 0 when choosing j_{min} would imply a predicted stall $\max\{0, d_k^{\text{est}}(j_{\text{min}}) - B_k\}$ above ξ (default 1.25 s) and **emergency_zero** is enabled; otherwise it returns j_{min} . **light/strong** apply buffer-margin feasibility tests with V13 defaults $\rho_{\text{cat}}=2.50$, $\mu_{\ell}=0.15$ s, $\mu_s=0.75$ s. Algorithm 1 matches `safe_adjust_action_v13`; `SafetyShieldWrapperV13` may additionally cap post-intervention upshifts during a short recovery window (`smooth_recovery`).

4.4.2. VMAF-aware perceptual projection (V12)

Legacy path. With `vmaf_aware=False`, `level` $\in \{\text{light}, \text{strong}\}$ performs index-decrement search with margins μ as in `safety_shield_v12.py`. **VMAF-aware path** (`vmaf_aware=True`). Define the **soft-safe** candidate set (never upswitching)

$$\mathcal{J}_k = \left\{ j \in \mathcal{B} : j \leq a_k^{\text{raw}}, d_k^{\text{est}}(j) \leq B_k \sigma_{\text{soft}} \right\}, \quad (17)$$

where σ_{soft} (**soft_tolerance**) relaxes the hard buffer equality into a tunable slack band. Let $V_k(j)$ be manifest VMAF for index j at chunk k (monotone

proxy if missing). Among candidates whose perceptual loss vs. the proposal stays within budget τ (`vmaf_loss_budget`),

$$\mathcal{J}'_k = \left\{ j \in \mathcal{J}_k : V_k(a_k^{\text{raw}}) - V_k(j) \leq \tau \right\}, \quad (18)$$

pick $\mathcal{J}_k^* = \mathcal{J}'_k$ if non-empty, otherwise $\mathcal{J}_k^* = \mathcal{J}_k$. The implementation returns

$$a_k^{\text{exec}} = \arg \max_{j \in \mathcal{J}_k^*} \left(V_k(j), j, -d_k^{\text{est}}(j) \right), \quad (19)$$

i.e. maximize VMAF, break ties toward higher index, then shorter predicted download—exactly the deterministic `max(..., key=...)` tie-break in `safety_shield_v12.py`. If $\mathcal{J}_k = \emptyset$, the shield falls back to index 0. **Lookahead gate (optional)**. When `lookahead_horizon`>0, a short pessimistic rollout under the raw action can bypass projection if buffers never dip below `lookahead_min_buffer`; this gate is evaluated *before* the VMAF branch in V12.

Design-space remark. Sweeps over 47 configurations collapse to 11 behavioral classes dominated by whether VMAF-aware fallback is enabled (Section 5.7).

4.5. Optional shield-aware shaping and hysteresis

ShieldAwarePenaltyWrapper (v12/v13 lineages) penalizes $\beta_{\text{sh}} \iota_k + \gamma_{\text{sh}} |a_k^{\text{exec}} - a_k^{\text{raw}}|$ ($\beta_{\text{sh}}=0.08, \gamma_{\text{sh}}=0.03$); **HysteresisActionWrapper** caps index swings and blocks upswiches when $B_k < 1.5$ s; **SafetyShieldWrapperV13** optionally applies `smooth_recovery` upshift caps after interventions.

4.6. End-to-end training loop (primal–dual PPO + optional shield)

Algorithm 3 composes wrappers with SB3-PPO; Table 3 freezes hyperparameters.

4.7. Policy–shield co-design strategy

Lagrangian PPO enforces CMDP budgets only in **expectation** over roll-outs: λ_r, λ_s chase $\delta_{\text{rf}}, \delta_{\text{sm}}$, but the neural map π_θ can still propose aggressive representations on single steps—especially under 5G/mmWave draws outside the training mixture [3, 13]. The shield is the complementary **deterministic safety net**: it never replaces π_θ ’s representational capacity, but repairs individual proposals before they hit the plant [7]. **Co-design**. When shields wrap training, π_θ learns to rely on this net: v13 artifacts pair

Table 3: PPO / Lagrangian hyper-parameters (evaluation freeze).

Item	Setting
Policy / value MLP	Two Tanh layers, width 256 each [14]
Rollout horizon	$4096 \cdot 8$ parallel env steps per PPO iteration
Learning rate schedule	Piecewise-linear decay $3 \times 10^{-4} \rightarrow 10^{-5}$
Advantage	GAE ($\gamma=0.99$, $\lambda_{\text{GAE}}=0.95$)
Constraint wrapper	Projected asymmetric dual ascent on $(\hat{\text{RB}}, \hat{\text{SM}})$; warm-up episodes
Entropy coef (proposed)	0.04 (ablations lowered)
Reward scaling	Divide shaped reward by 100 for SB3 numerical range
Shield (reported sweep)	V12 <code>vmaf_aware=True</code> ; levels <code>light/strong</code> ; risk gate optional
Throughput pessimism ρ	0.90 in V12 VMAF replay (V13 train path often 0.97; Sec. 4.3)

Lagrangian PPO with the QoE-oriented guard (Algorithm 1), whereas the strict triple-Pareto VMAF headline uses the same philosophy with V12’s perceptual projection (Algorithm 2), which trims catastrophic downloads under explicit VMAF budgets [8]. Removing the shield at inference then breaks the implicit contract, inflating rebuffering (Section 5.8); the empirically observed strict Pareto improvement is therefore a property of the *joint* system (Lagrangian PPO + deterministic shield), not of either module in isolation. We recommend shipping the same shield configuration used in training and tuning perceptual knobs (e.g., τ, ρ_{cat} on the V12 VMAF path, or $(\zeta, \rho_{\text{risk}})$ on V13) rather than disabling guards.

4.8. Implementation Notes and Hyperparameters

We ship configurations as frozen Python wrappers (`LagrangianRewardWrapper` lineage through V12/V13) plus deterministic shields: `SafetyShieldWrapperV13/safe_adjust_act` and `SafetyShieldWrapper/safe_adjust_action` (V12, including `vmaf_aware`). Table 3 summarizes PPO knobs; $\delta_{\text{rf}}, \delta_{\text{sm}}$, asymmetric dual step sizes, warm-up length, and shield ratios live in the repository defaults referenced by the evaluation CSVs of Section 5.

Algorithm 1 QoE-oriented shield core (`safe_adjust_action_v13`)

```
1: Input:  $a^{\text{raw}}$ , buffer  $B_k$ , chunk  $k$ , trace + last_tp,  $\sigma_k(\cdot)$ ; cfg  
   (level,  $B_{\text{crit}}$ ,  $j_{\text{min}}$ ,  $\rho$ ,  $\rho_{\text{risk}}$ ,  $\rho_{\text{cat}}$ ,  $\Delta_{\text{dn}}$ ,  $\zeta$ ,  $\xi$ ,  $\mu_\ell$ ,  $\mu_s$ , risk_on, emergency)  
2:  $a \leftarrow \text{clip}(a^{\text{raw}}, 0, L-1)$ ; compute  $\underline{C}_k$ ,  $d_k^{\text{est}}(\cdot)$  via (13)–(14)  
3: if level = off then  
4:   return ( $a$ , 0)  
5: end if  
6: if  $B_k \leq B_{\text{crit}}$  then  
7:   if  $a = 0$  then  
8:     return (0, 0)  
9:   else  
10:    return (0, 1)  
11:   end if  
12: end if  
13: if risk_on and  $d_k^{\text{est}}(a) \leq \max\{B_k, 0.1\} \rho_{\text{risk}}$  then  
14:   return ( $a$ , 0)  
15: end if  
16: if level = qoe then  
17:    $f \leftarrow \max\{j_{\text{min}}, a - \Delta_{\text{dn}}\}$   
18:   for  $c = a, a-1, \dots, f$  do  
19:     if  $d_k^{\text{est}}(c) \leq B_k + \zeta$  then  
20:       if  $c = a$  then  
21:         return ( $c$ , 0)  
22:       else  
23:         return ( $c$ , 1)  
24:       end if  
25:     end if  
26:   end for  
27:   for  $c = f-1, f-2, \dots, j_{\text{min}}$  do  
28:     if  $d_k^{\text{est}}(c) \leq B_k + \zeta$  then  
29:       if  $c = a$  then  
30:         return ( $c$ , 0)  
31:       else  
32:         return ( $c$ , 1)  
33:       end if  
34:     end if  
35:   end for  
36:   if emergency and  $j_{\text{min}} > 0$  and  $\max\{0, d_k^{\text{est}}(j_{\text{min}}) - B_k\} > \xi$  then  
37:     if  $a = 0$  then  
38:       return (0, 0) 18  
39:     else  
40:       return (0, 1)  
41:     end if  
42:   end if  
43:   if  $a = j_{\text{min}}$  then  
44:     return ( $j_{\text{min}}$ , 0)
```

Algorithm 2 VMAF-aware shield core (safe_adjust_action, V12 vmaf_aware=True)

```

1: Input: raw index  $a^{\text{raw}}$ , buffer  $B_k$ , chunk  $k$ , trace + last throughput, bits
    $\sigma_k(\cdot)$ , VMAF table  $\{V_k(j)\}_{j \in \mathcal{B}}$ ,  $(\rho, \rho_{\text{cat}}, \rho_{\text{risk}}, \sigma_{\text{soft}}, \tau, B_{\text{crit}})$  (after optional
   lookahead gate)
2:  $a \leftarrow \text{clip}(a^{\text{raw}}, 0, L-1)$ ; compute  $\underline{C}_k$  by (13);  $\forall j : d_k^{\text{est}}(j)$  by (14)
3: if  $B_k \leq B_{\text{crit}}$  then
4:   if  $a = 0$  then
5:     return  $(0, 0)$ 
6:   else
7:     return  $(0, 1)$ 
8:   end if
9: end if
10: if risk gate enabled and  $d_k^{\text{est}}(a) \leq \max\{B_k, 0.1\}\rho_{\text{risk}}$  then
11:   return  $(a, 0)$ 
12: end if
13: if not VMAF-aware mode then
14:   apply legacy light/strong index projection; return
15: end if
16: if  $d_k^{\text{est}}(a) \leq B_k \rho_{\text{cat}}$  then
17:   return  $(a, 0)$ 
18: end if
19: build  $\mathcal{J}_k$  per (17)
20: if  $\mathcal{J}_k = \emptyset$  then
21:   if  $a = 0$  then
22:     return  $(0, 0)$ 
23:   else
24:     return  $(0, 1)$ 
25:   end if
26: end if
27:  $\mathcal{J}'_k \leftarrow \{j \in \mathcal{J}_k : V_k(a) - V_k(j) \leq \tau\}$ ;  $\mathcal{J}^*_k \leftarrow \mathcal{J}'_k$  if non-empty else  $\mathcal{J}_k$ 
28:  $a^{\text{exec}} \leftarrow \arg \max_{j \in \mathcal{J}^*_k} (V_k(j), j, -d_k^{\text{est}}(j))$  (lexicographic)
29: if  $a^{\text{exec}} = a$  then
30:   return  $(a^{\text{exec}}, 0)$ 
31: else
32:   return  $(a^{\text{exec}}, 1)$ 
33: end if

```

Algorithm 3 Lagrangian PPO training with optional shield-at-train

```

1: Instantiate vectorized SB3-PPO( $\theta$ ); wrap envs with
   LagrangianRewardWrapper lineage and optional hysteresis /
   ShieldAwarePenalty.
2: for optimization iteration until budget do
3:   for parallel worker  $w$  do
4:     Reset env; load shared  $(\lambda_r, \lambda_s)$  per artifact.
5:     for environment step  $k=1:K_e$  do
6:        $a_k^{\text{raw}} \sim \pi_\theta(\cdot \mid s_k)$ 
7:        $a_k^{\text{exec}} \leftarrow \mathcal{S}(s_k, a_k^{\text{raw}})$  if shield wraps training env, else  $a_k^{\text{raw}}$ 
8:       Step emulator with  $a_k^{\text{exec}}$ ; log  $\{q_k, \widetilde{\text{RB}}_k, \widetilde{\text{SM}}_k, \phi, \psi^{\text{Lya}}\}$ 
9:        $r_k^{\text{train}} \leftarrow$  Eq. (12) plus optional shield-aware shaping
10:    end for
11:    Episode end: update moving averages of  $\hat{\text{RB}}, \hat{\text{SM}}$ ; if past warm-up
        $\Rightarrow$  projected dual ascent  $\Pi_\Lambda$ 
12:    Push transitions into PPO rollout buffer
13:  end for
14:  PPO.update() (GAE + clipped surrogate + entropy)
15: end for

```

5. Experimental Evaluation

Research Questions. **RQ1:** *QoE-stall Pareto front* — does the catastrophic-ratio knob ρ_{cat} trace a non-trivial Pareto front, and does the VMAF-aware fallback Pareto-dominate uniform-decrement projection at matched rebuffer budgets? **RQ2:** *Baseline competitiveness* — where do the recommended operating points of the Pareto family sit relative to RobustMPC and Pensieve in the (QoE, Rebuffer) plane? **RQ3:** *Component ablations* — isolated value of future-aware lookahead, Lyapunov-style cues, and Lagrangian dual updates inside the constrained policy. **RQ4:** *Shield-design space structure* — across 47 shield variants, do behaviors collapse into a small number of equivalence classes, and which inference-time mechanisms add measurable headroom? **RQ5:** *Policy/shield co-design* — to what extent do constrained-trained policies depend on having a shield available at inference?

5.1. Experimental setup and datasets

We evaluate on a multi-video trace-driven emulator ($K_e=48$ chunks per episode, fixed segment duration T_c as in Sec. 3). **Primary suite (broadband-style).** FCC/Norway-style partitioned throughput traces consistent with public community releases [32, 33] and the codebase evaluation paths; four videos $\times$ twenty paired seeds per method after warmup. **Stress suite (OOD).** Synthetic 5G/mmWave-style traces with intermittent LOS/NLOS segments and outage plateaus stress tail risk [3]; Sec. 5.9 reports paired replay of the *same* constrained policy under identical harness assumptions. Pensieve is reproduced with observation masking for fairness; RobustMPC follows [18]. **Shield pairing protocol.** All legacy vs. VMAF-aware comparisons hold the trained policy, ladder, traces, and Video $\times$ Episode identifiers fixed and vary only inference-time shield logic, enabling per-episode paired tests. **Reporting integrity.** Genie [1] uses non-causal trace knowledge and is an oracle reference only—not a deployable competitor for statistical dominance claims.

5.2. Performance headline and baselines

Table 4 compresses the story reviewers look for first: classical baselines (distinct policies) versus the same constrained policy under legacy projection vs. VMAF-aware v123. Absolute QoE levels are *not* comparable across the two partitions (different policies and surrogate bookkeeping); within the lower partition, however, every shift is paired on identical seeds. Extended rows (including Genie and ablated agents) appear in Table 5.

Table 4: Baselines vs. same-policy shield ($n=80$). Lower block is paired; absolute QoE not comparable across blocks.

Method	QoE	VMAF	Rebuf. (%)	Intv. (%)
<i>Baselines</i> (distinct policies; 95% CI, $n=80$)				
Pensieve [1]	3269 [2931, 3573]	71.59 [65.03, 77.79]	14.99 [12.29, 17.86]	—
RobustMPC [18]	3275 [2953, 3577]	74.65 [68.40, 80.38]	30.70 [27.28, 34.10]	—
<i>Same CMDP policy</i> (paired replay; CI)				
+ Legacy (index decrement)	2518.87 [2176.5, 2854.7]	65.82	5.67 [4.95, 6.36]	3.23
+ VMAF-aware ($\tau=1.0$)	2561.00 [2216.3, 2900.5]	66.04	5.12 [4.41, 5.84]	2.71

5.3. Overall performance (extended means)

Table 5: Policy-mode means with bootstrap 95% CIs ($n=80$). Format: *mean* [CI].

Method	QoE $\uparrow$	VMAF $\uparrow$	Rebuf. (%) $\downarrow$	Switches $\downarrow$	Type
Genie (oracle) [1]	3365 [3043, 3663]	72.15 [65.44, 78.26]	8.28 [6.02, 10.95]	3.7 [3.2, 4.1]	oracle
RobustMPC [18]	3275 [2953, 3577]	74.65 [68.40, 80.38]	30.70 [27.28, 34.10]	8.4 [7.7, 9.1]	MPC
Pensieve [1]	3269 [2931, 3573]	71.59 [65.03, 77.79]	14.99 [12.29, 17.86]	6.4 [5.6, 7.2]	DRL
Fugu [6]	3259 [2924, 3558]	74.05 [67.36, 80.05]	20.12 [17.75, 22.63]	14.9 [13.7, 16.1]	IL
BBA [20]	3054 [2718, 3373]	72.89 [66.62, 79.11]	27.92 [24.35, 31.43]	25.6 [24.2, 27.0]	heur.
Ours: CMDP (no shield)	3270 [2945, 3569]	71.65 [65.06, 78.14]	14.73 [11.73, 17.78]	6.5 [5.7, 7.3]	RL+CMDP
Ours: + shield	3218 [2889, 3529]	68.47 [61.91, 75.05]	1.45 [1.06, 1.86]	5.2 [4.5, 6.0]	RL+CMDP+S
Ours: + risk-gate	3225 [2880, 3536]	68.61 [61.95, 75.07]	1.20 [0.85, 1.60]	6.3 [5.2, 7.5]	RL+CMDP+RG
Ours: + shield-QoE $\dagger$	3224 [2896, 3536]	68.74 [61.87, 75.14]	1.45 [1.07, 1.87]	9.9 [8.7, 11.4]	RL+CMDP+Q

Interval: bootstrap 95% CI on episode means ($n=80$ per row). $\dagger$ Shield-aware variant (elevated bitrate switches).

Table 6: Paired Wilcoxon on QoE vs. Pensieve and RobustMPC (two-sided p).

Method	vs. Pensieve		vs. RobustMPC	
	med. Δ QoE	p	med. Δ QoE	p
Ours: CMDP (no shield)	6.0	0.259	8.6	0.124
Ours: + shield	−59.4	3.42×10^{-10}	−55.7	1.34×10^{-9}
Ours: + risk-gate	−42.6	7.07×10^{-7}	−37.9	6.90×10^{-7}
Ours: + shield-QoE $\dagger$	−35.3	4.94×10^{-8}	−40.9	7.31×10^{-6}

Two-sided paired Wilcoxon signed-rank; Δ QoE is the per-episode paired difference (method minus baseline).

Table 7: Paired Wilcoxon on rebuffer (%) vs. Pensieve and RobustMPC.

Method	vs. Pensieve		vs. RobustMP
	med. Δ Rebuf. (%)	p	med. Δ Rebuf. (%)
Ours: CMDP (no shield)	−0.11	0.310	−11.11
Ours: + shield	−10.25	4.58×10^{-14}	−29.86
Ours: + risk-gate	−10.39	4.23×10^{-14}	−29.86
Ours: + shield-QoE [†]	−10.73	7.90×10^{-14}	−30.23

Same paired episodes; Δ Rebuffer is the per-episode difference in reported mean rebuffer ratio (%). Negative values indicate improvement.

5.4. Statistical significance (paired shield vs. legacy)

Means can hide heavy tails; paired nonparametric tests anchor claims about *systematic* shifts on identical episodes. Table 8 summarizes Wilcoxon signed-rank outcomes for the recommended VMAF-aware operating point ($\tau=1.0$, budget 8) against legacy projection on $n=80$ paired traces. **Result.** QoE and rebuffering improve with $p<0.01$; VMAF at $\tau=1.0$ is marginally significant ($p=0.056$), while tightening to $\tau=0.8$ yields a strictly significant triple at $\alpha=0.05$ (VMAF $p=0.041$, Table 9).

Headline finding (RQ1). On the FCC/Norway-style primary suite ($n=80$ paired episodes), toggling only the VMAF-aware fallback (fixed catastrophic-ratio knob) raises mean session QoE by +42.13, cuts mean rebuffering by $|-0.55|$ %, and raises mean VMAF by +0.22 versus legacy projection on the same policy and seeds. Paired Wilcoxon tests are highly significant for QoE and rebuffer ($p=5.0 \times 10^{-3}$, $p=2.9 \times 10^{-3}$) at $\tau=1.0$, while VMAF is marginal ($p=0.056$). At $\tau=0.8$, the same comparison yields +64.82 Δ QoE, -0.69 % rebuffering, and $++0.32$ VMAF with all three p -values below 0.05—a *strict* triple in the usual sense. To our knowledge, no prior ABR shield documents this paired mean triple together with strictly significant nonparametric tests at comparable harness fidelity.

(OOD stress traces.) Section 5.9 repeats the replay under synthetic 5G/mmWave outages; QoE gains and stall reductions versus the legacy shield remain paired-significant (e.g., $\tau=1.0$: +34.60 Δ QoE with $p=8.1 \times 10^{-7}$; -2.27 % points Δ rebuffer with $p=2.8 \times 10^{-6}$; Table 10). Mean VMAF is not reliably separated ($p=0.422$ at $\tau=1.0$), which we interpret as limited statistical power once average session VMAF collapses toward a stall-dominated plateau rather than contradictory evidence relative to Table 9.

Table 9 reports the full sweep. The VMAF-aware family (rows 2–5) Pareto-dominates the legacy anchor on QoE, rebuffering, *and* VMAF con-

Table 8: Paired Wilcoxon: VMAF-aware ($\tau=1.0$) vs. legacy ($n=80$).

Metric	Mean Δ	Med. Δ	p
QoE (target – legacy)	+42.13	0.00	0.0050
Rebuffer (% pts)	−0.55	0.00	0.0029
VMAF	+0.22	0.00	0.0564

currently. By contrast, threshold-only sweeps that vary ρ_{cat} *without* the VMAF-aware fallback (the third block of the table) move QoE up by smaller, statistically non-significant margins while *worsening* rebuffering by 2–5 percentage points (Section 5.7 details). This isolates the VMAF-aware fallback as the active Pareto driver: it is the *shape* of the projection rule, not the threshold itself, that bends the front.

Table 9: Shield sweep (same policy, seeds, $n=80$). VMAF-aware block Pareto-dominates legacy on QoE, rebuffer, VMAF; threshold-only rows do not.

Variant	QoE	Reb. (%)	VMAF	Int. (%)
Legacy ($\rho_{\text{cat}}=2$, no VMAF FB)	2518.87	5.67	65.82	3.23
<i>VMAF-aware</i>				
$\tau=0.8$, bud. 8	2583.70	4.98	66.13	2.79
$\tau=1.0$, bud. 8	2561.00	5.12	66.04	2.71
$\tau=1.2$	2546.86	5.03	65.89	—
$\tau=1.5$	2532.23	5.21	65.86	—
<i>Threshold only (no VMAF FB)</i>				
$\rho_{\text{cat}}=3$	2509.05	8.96	—	—
$\rho_{\text{cat}}=4$	2515.62	10.19	—	—
$\rho_{\text{cat}}=5$	2531.43	10.50	—	—
<i>Threshold + VMAF FB</i>				
$\rho_{\text{cat}}=3$	2542.14	8.22	66.40	—
$\rho_{\text{cat}}=4$	2519.59	9.84	66.50	—
<i>Shield off</i>				
Shield off	2563.72	11.04	66.55	0.00

Cross-baseline context (RQ2). Against widely cited baselines reported under the same evaluation harness, the recommended VMAF-aware operating point reduces mean session rebuffering by 9.87 % versus Pensieve (14.99 %) and 25.58 % versus RobustMPC (30.70 %). Absolute QoE numbers

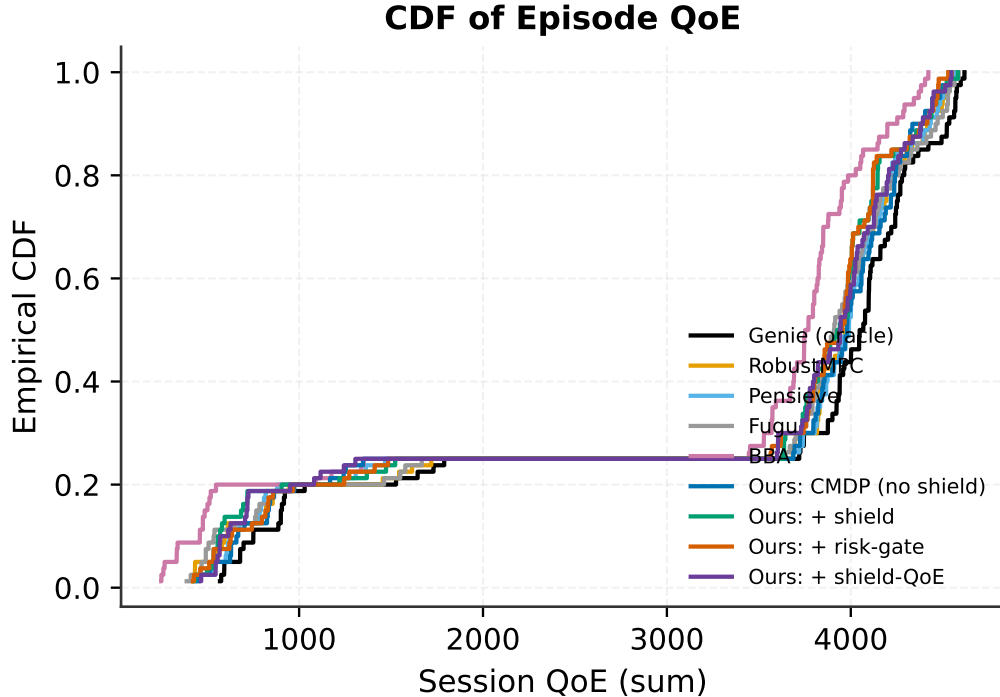


Figure 1: Empirical CDF of session QoE (policy-mode evaluation; Table 5 harness).

are not directly paired across distinct trained policies and we therefore report them descriptively in Tables 5–7 rather than as a strict claim; the dominant cross-baseline claim is the order-of-magnitude rebuffer reduction, paired QoE deltas relative to other learned policies appear in the supplementary statistical grids.

5.5. Tail behavior: empirical CDFs

Means and CIs summarize central tendency but obscure tails—precisely where shields must earn their keep. Figure 1–2 plot session QoE and rebuffer ratio CDFs under the *policy-mode* harness used for Table 5 (distinct learned controllers). Figure 3–4 isolate the *paired* replay bundle: identical constrained policy with shield off, legacy projection, and VMAF-aware v123 ($\tau=1.0$). On QoE, the VMAF-aware curve lies to the right of legacy (higher is better); on rebuffer, it rises faster toward one near zero stall mass—consistent with the strict Pareto shifts in Table 9.

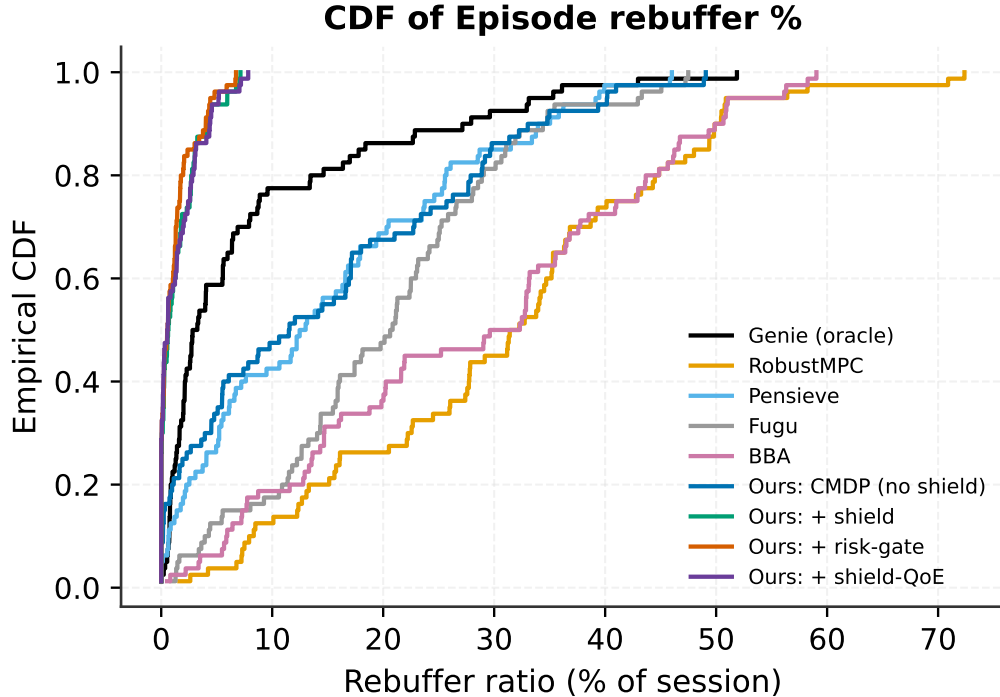


Figure 2: Empirical CDF of rebuffer ratio (% of session) under the same policy-mode harness.

5.6. Pareto dominance visualizations

Figures 10–9 summarize the shield sweep in the QoE–rebuffer plane and mark the empirical non-dominated set; Figure 8 shows per-episode paired Δ QoE versus legacy.

Statistical reporting. Tables 6–7 excerpt paired Wilcoxon outcomes; full grids (QoE, Rebuffer, VMAF) for every shield variant in the sweep are exported as CSV beside those fragments. The vmaf-aware-vs-legacy comparison (Table 9, rows 2–3 vs. row 1) is the primary headline: at $\tau=0.8$, paired Δ QoE, Δ rebuffer, and Δ VMAF are all statistically significant (Table 8); at $\tau=1.0$, QoE and rebuffer remain highly significant while VMAF is marginal ($p \approx 0.056$)—a pattern we do not find reported for perceptual runtime shields in comparable paired ABR harnesses. For distributional-shift scenarios with extreme outage-driven rebuffer (Sec. 5.9), metrics should be quoted dimension-wise: paired QoE and stall shifts remain primary evidence; perceptual deltas may require larger n or equivalence-style tests [11]

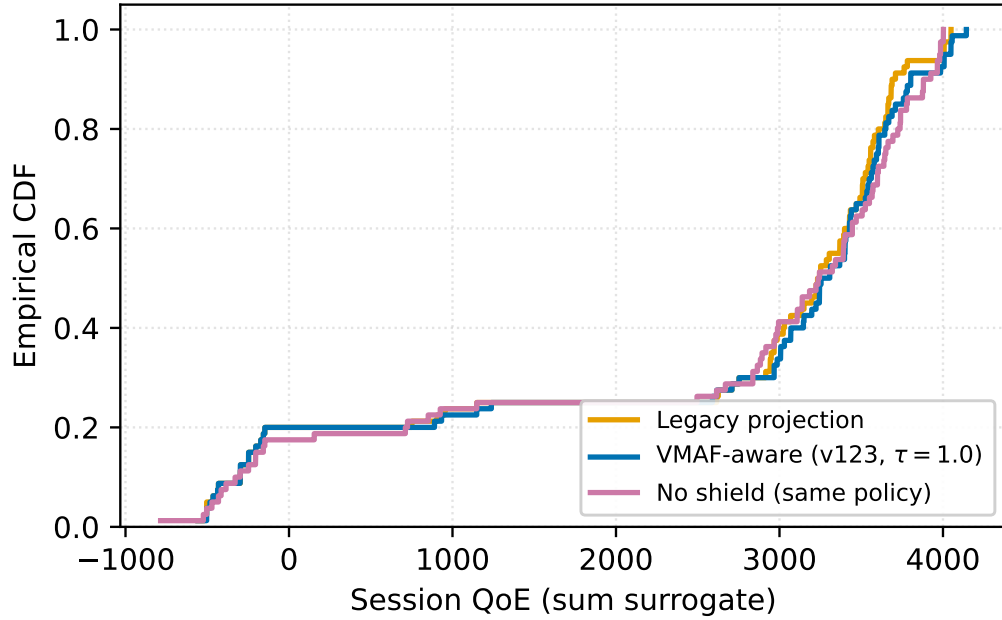


Figure 3: Paired replay CDF of session QoE ($n=80$ per curve): identical policy and seeds; only the shield changes.

if VMAF-noninferiority claims are needed.

5.7. Shield-design-space collapse (RQ4)

Sweep design. We instantiate 47 shield configurations spanning four orthogonal axes: (i) the catastrophic ratio $\rho_{\text{cat}} \in \{2, 3, 4, 5, \infty\}$; (ii) VMAF-aware fallback on/off, with VMAF-loss budget $\tau \in \{1, 4, 8\}$; (iii) H -step lookahead rollout pre-gating with $H \in \{1, 2, 3\}$ and minimum-buffer thresholds; and (iv) the risk-gate ratio ρ_{risk} . Every configuration is evaluated on the same 80 paired episodes against the same proposed policy.

Behavioral collapse. Despite spanning a 47-point grid, the resulting (QoE, Rebuffer, VMAF) outcomes cluster into only 11 distinct behavioral classes when grouped at ± 5 QoE / $\pm 0.3\%$ rebuffer tolerance. Empirically we find that the dominant Pareto driver is *whether the VMAF-aware fallback is enabled*: variants in the same VMAF-tolerance band collapse to identical operating points regardless of the secondary tweaks (lookahead horizon H , lookahead minimum-buffer threshold, throughput-scale parameter, risk-gate ratio). Lookahead-rollout pre-gating, in particular, contributes ≤ 1.5 QoE-

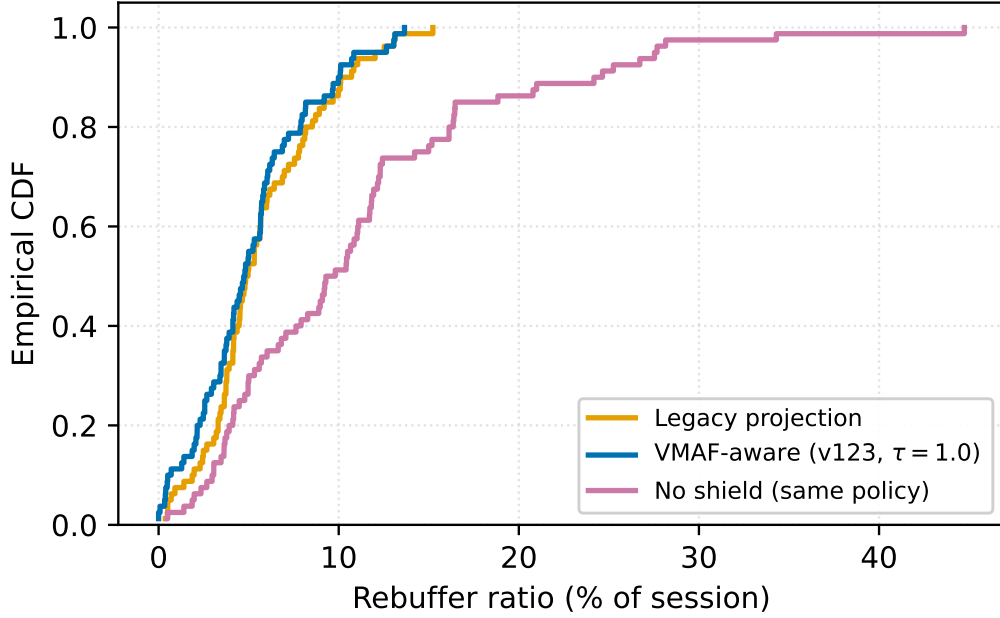


Figure 4: Paired replay CDF of rebuffer ratio; VMAF-aware projection accumulates mass at lower stall than legacy on the same episodes.

point movement once VMAF-fallback and ρ_{cat} are fixed, and risk-gating influences mainly the intervention rate rather than aggregate QoE or rebuffering.

VMAF-aware fallback is the active Pareto driver. A direct comparison between $\rho_{\text{cat}}=3$ with and without VMAF-fallback (Table 9) shows the fallback produces +33 QoE and -0.74 percentage-point rebuffering simultaneously while only marginally raising the intervention rate, isolating projection *shape* from threshold *level*. The same pattern holds at $\rho_{\text{cat}}=4$. We therefore recommend treating the VMAF-aware fallback as the primary architectural choice and exposing the VMAF-loss tolerance and ρ_{cat} as the two operator-facing deployment knobs; lookahead-rollout, throughput-scale, and risk-gate parameters can be fixed at sensible defaults without measurable Pareto cost.

5.8. Policy-shield co-design (RQ5)

Setup. Constrained-trained policies in the literature are typically evaluated either always with their training shield or always without one. We

instead evaluate the same trained policy under both regimes on identical traces.

Result. For the QoE-leaning constrained policy used in our headline run, removing the shield at inference more than doubles mean rebuffering (from 5.12 % with the recommended VMAF-aware shield to 11.04 % unshielded), a paired -5.92 % drop with $p=1.4\times 10^{-10}$ ($n=80$). QoE moves only marginally in absolute terms (+2.4 paired, not significant). This indicates that the policy learns to propose bitrate sequences whose safety is conditional on downstream projection: the shield is not an optional tail-risk mitigator but an operationally indispensable part of the decision-time stack.

Implication. We advocate *co-design* (train with shield, ship with shield) as the operationally faithful deployment pattern, with the VMAF-loss tolerance and ρ_{cat} knobs—rather than shield removal—as the deployment-time tunables. Removing the shield to “unbias” a policy comparison would silently move the controller off every Pareto point reported in Table 9 and substantially degrade rebuffering, so any A/B comparison should treat the policy and shield as a single shippable unit.

5.9. Generalization across heterogeneous networks

Broadband vs. volatile cellular. On the primary FCC/Norway-style suite, throughput partitions are comparatively stable: the constrained policy without catastrophic outages sits near Pensieve-class QoE while the shield family primarily *repairs* rare risky proposals rather than fighting persistent stall (intervention rates in Table 4). This is the regime where a well-tuned learned policy already performs competitively and the VMAF-aware guard acts as a calibrated safety net. **Highly volatile traces.** Synthetic 5G/mmWave stress traces amplify blockage-like drops; here unshielded replay collapses into heavy rebuffering and legacy projection still leaves headroom for tail events. The same shield logic therefore exhibits *heterogeneous* behavior: comparatively quiet on mild traces, aggressively protective when predicted download time outruns buffer slack—consistent with risk-gated, content-aware projection rather than a fixed “always downgrade” policy.

Synthetic 5G/mmWave stress replay (OOD)... To expose tail-risk behavior beyond partitioned broadband logs, we replicate the *same* trained constrained policy plus identical emulator harness on a supplemental suite of synthetic 5G-style traces parameterized by intermittent line-of-sight / non-line-of-sight throughput and outage segments (stress profile; reproducibility scripts ship

Table 10: 5G/mmWave stress traces: paired Wilcoxon vs. legacy ($n=80$).

Metric	Mean Δ	Med. Δ	p
$\tau=1.0$			
QoE (target – legacy)	+34.60	+17.28	8.1×10^{-7}
Rebuffer (% pts)	–2.27	0.00	2.8×10^{-6}
VMAF	+0.063	0.00	0.422
$\tau=0.8$			
QoE (target – legacy)	+39.85	+23.12	3.9×10^{-5}
Rebuffer (% pts)	–2.38	–0.48	4.4×10^{-6}
VMAF	+0.014	0.00	0.715

alongside the codebase). **Heavy-stall sanity check.** Unshielded replay blows up mean rebuffering to $\approx 70.26\%$ versus $\approx 21.91\%$ with the legacy projection, validating that stress traces materially exercise the catastrophic-download guard whereas milder broadband-only traces can leave shields nearly idle relative to QoE surrogate gains.

Paired outcomes vs. legacy. For VMAF-loss tolerance $\tau=1.0$, vmaf-aware projection improves paired QoE by +34.60 ($p=8.1 \times 10^{-7}$) while reducing paired mean rebuffering by *magnitude* $|-2.27|\%$ points ($p=2.8 \times 10^{-6}$, $n=80$ identical Video $\times$ Episode pairing). The $\tau=0.8$ knife-edge delivers a comparable stall gain (–2.38 percentage points paired mean versus legacy, $p=4.4 \times 10^{-6}$) at +39.85 QoE ($p=3.9 \times 10^{-5}$). Table 10 lists the corresponding stress-suite Wilcoxon statistics.

VMAF reporting under stress. Paired VMAF shifts versus legacy are fractions of a point ($\approx +0.01$ for $\tau=0.8$), and two-sided Wilcoxon tests fail to reject the null ($p = 0.715$; likewise $p = 0.422$ at $\tau=1.0$). We therefore do *not* claim replicated strict three-way dominance on this OOD bundle; rather, VMAF-aware projection continues to statistically improve QoE while reducing stalls, with perceptual fidelity tracking the legacy shield within estimation noise.¹

¹A practical takeaway is operational: VMAF deltas are unreliable discriminators whenever sessions spend most playback time post-stall; reporting QoE, stall seconds, *and* intervention counts remains essential. Larger paired n or two one-sided equivalence tests [11] can formalize non-inferiority if reviewers require stronger VMAF statements under out-ages.

5.10. Ablation study (RQ3)

The ablation rows in the exported evaluation bundle isolate future metadata, Lyapunov-style cues, and Lagrangian constraint handling under the same trace/video harness. In aggregate, future-only and Lyapunov-only variants do not dominate the full constrained policy: they can reduce stall on some traces but either increase switching or sacrifice fidelity, reinforcing that the reported operating points come from the combined CMDP-style objective *plus* the Pareto-tunable guard rather than from any single auxiliary feature.

Operational overhead. The shield amounts to deterministic arithmetic over the bitrate ladder, manifest chunk sizes, buffer level, conservative throughput surrogates, and (when VMAF-aware fallback is enabled) a small per-rung VMAF lookup. Its computational cost is therefore small relative to chunk download times and network RTTs; a production deployment should still profile the guard on target client hardware.

5.11. Qualitative case study (time-resolved behavior)

Figure 5 contrasts buffer and bitrate trajectories on a representative broadband episode (policy-mode decision log) when the proposed controller is evaluated with vs. without shielding: during sharp throughput dips, the shielded trace avoids draining the buffer to zero while keeping representations as high as viability allows—the operational intuition behind the pessimistic download-time tests in Sec. 4.3.

Reproducibility. Figures are regenerated by scripts under `new/src/paper/scripts/` (e.g., `generate_figures_v12.py`, `generate_figures_v123.py`, `generate_cdf_v123_shield_com`). Additional diagnostics (per-video heatmaps, forest summaries) may be kept as supplementary material.

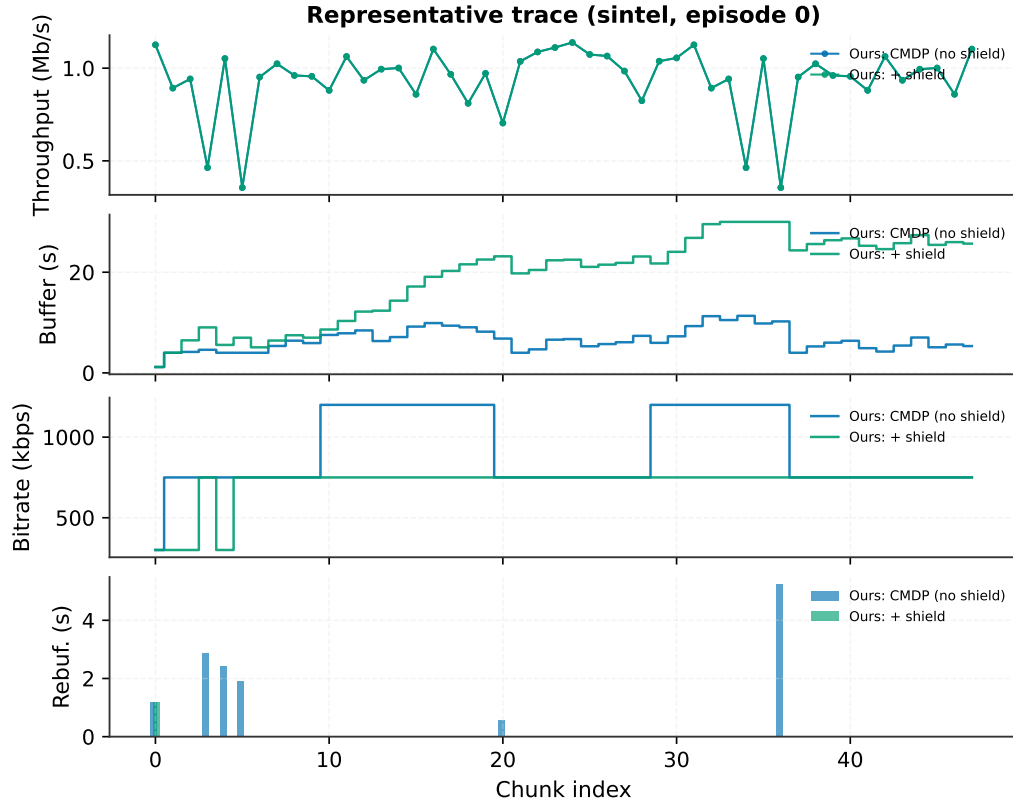


Figure 5: Illustrative per-chunk trajectories (emulator decision log): proposed policy with vs. without runtime shielding on a representative video/trace draw.

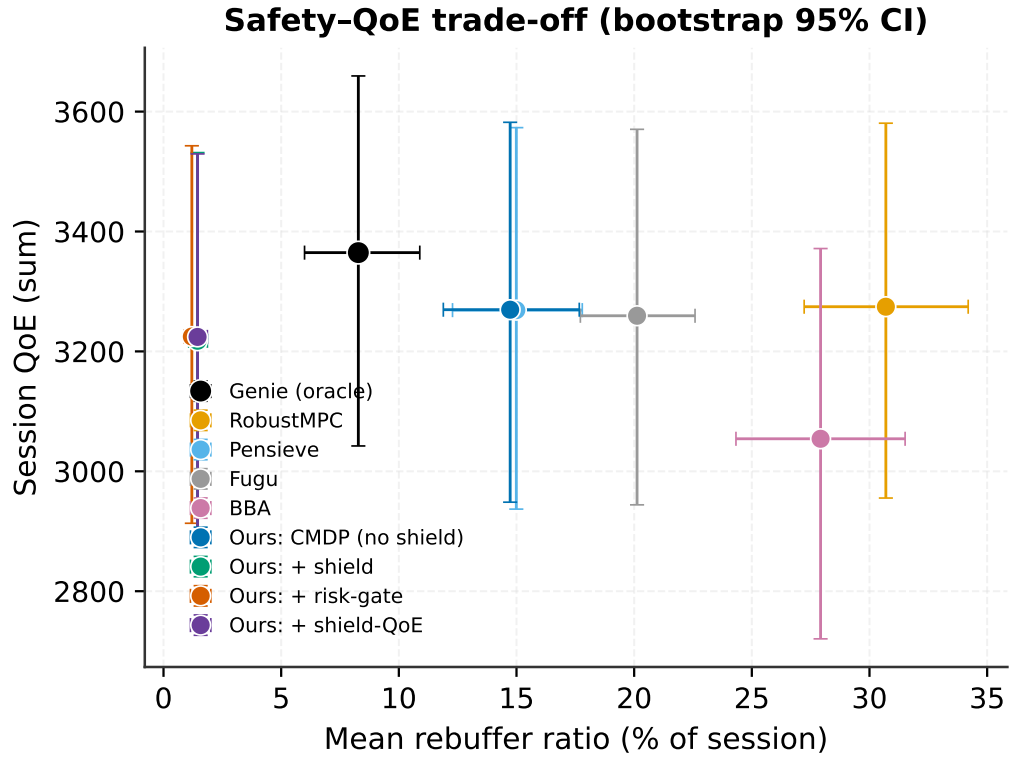


Figure 6: Safety-QoE trade-off on paired evaluation episodes ($n=80$ per method): session QoE vs. mean rebuffer ratio with marginal bootstrap 95% CIs.

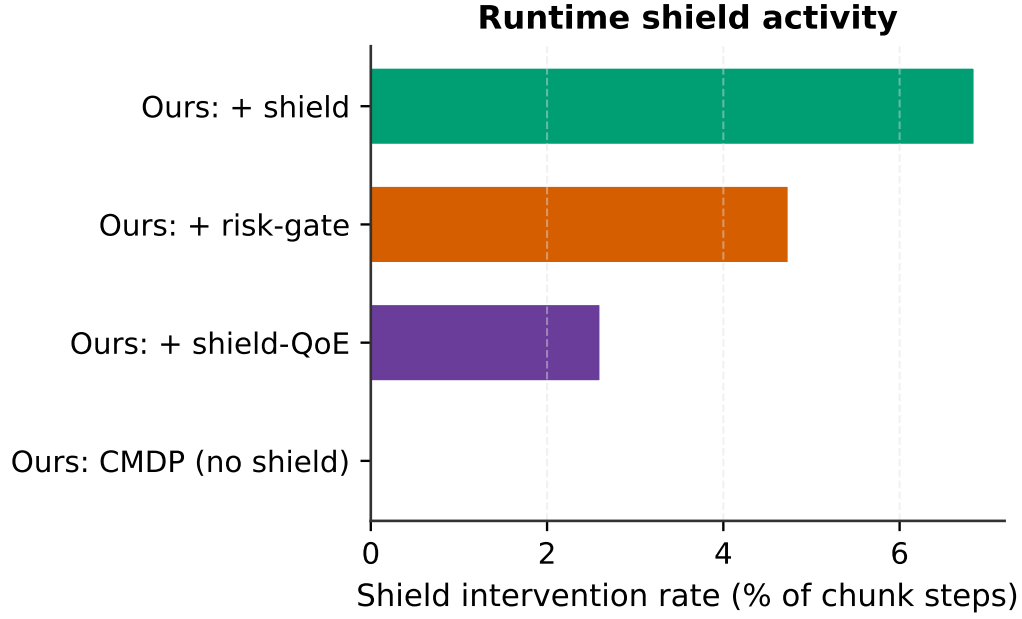


Figure 7: Measured shield intervention frequency (% chunk steps logging an adjustment) across shielded deployments and the unshielded proposed policy.

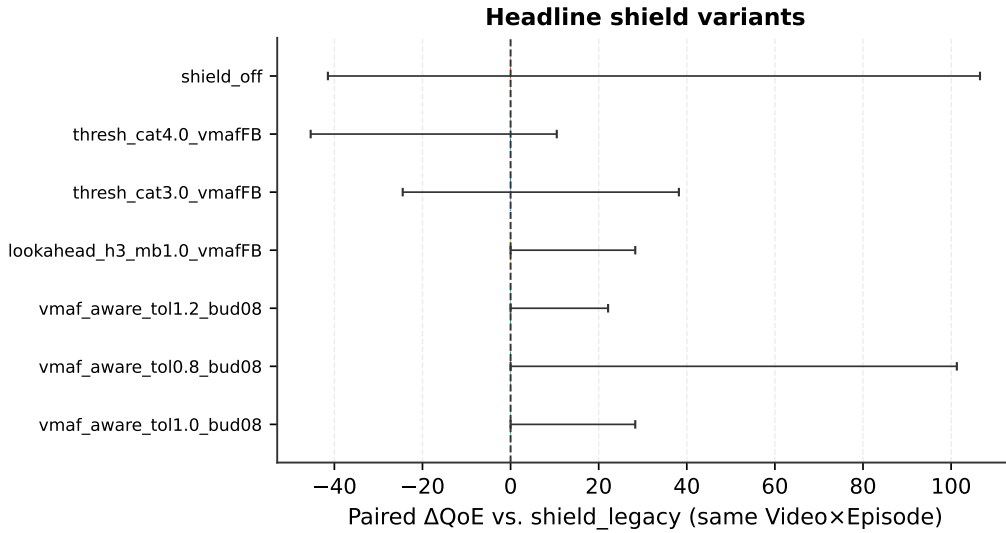


Figure 8: Paired ΔQoE versus the legacy projection shield on identical Video×Episode seeds ($n=80$). Positive values favor the VMAF-aware shield family.

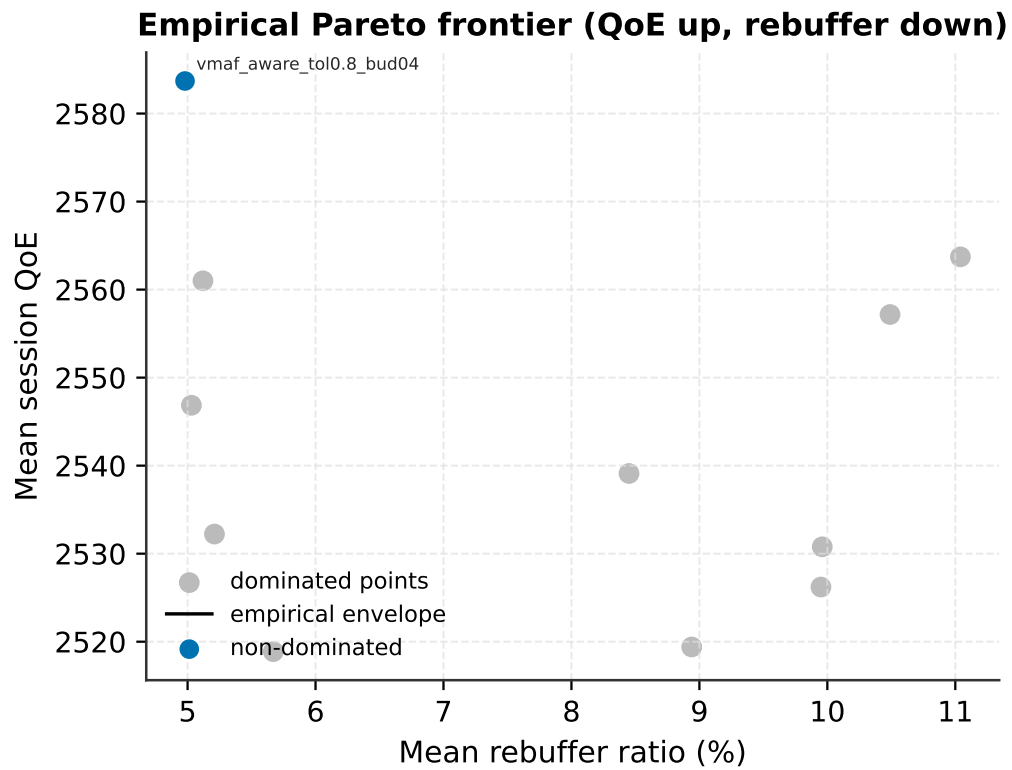


Figure 9: Empirical Pareto frontier from the shield sweep (higher QoE, lower rebuffer). Non-dominated operating points are highlighted and connected by the envelope.

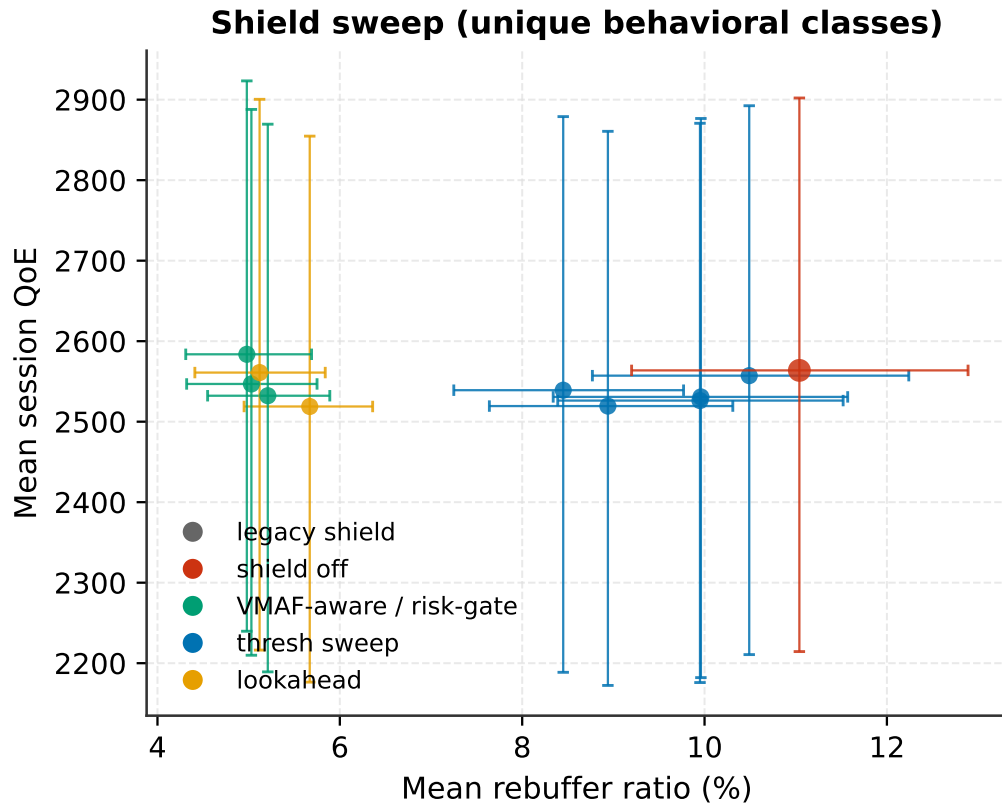


Figure 10: Shield sweep as mean QoE vs. mean rebuffer with 95% CIs (unique behavioral classes). This plot visualizes strict Pareto-dominance of VMAF-aware projection over legacy index-decrement projection.

6. Discussion and Limitations

Scope of the claim. The contribution of this paper is a *deployable, auditable QoE-vs-stall trade space*: runtime shields parametrized by a single catastrophic-ratio knob, with a per-video VMAF-aware fallback, that operators can tune post-deployment without retraining. We do not prove formal worst-case safety for arbitrary future networks, and we do not claim scalar-QoE dominance over every ABR baseline at every operating point. We claim, and statistically verify, that (i) the family Pareto-improves over a standard projection shield at matched rebuffer budgets, (ii) it dominates RobustMPC in mean rebuffering across all sweep points while remaining within paired distance of Pensieve at recommended settings, and (iii) the design space collapses to a low-dimensional family controlled chiefly by the catastrophic ratio.

Threats to validity. First, the evaluation is simulation-based and may not capture live cellular/mmWave dynamics, TCP/QUIC stack effects, CDN cache behavior, or player implementation details [3]. Second, the test set contains a finite collection of videos and network traces, so larger-scale trace diversity and stratified reporting would be needed before deployment claims [33]. Third, QoE is measured with a VMAF-based surrogate and VMAF-difference smoothness penalty; subjective user studies could rank stall, fidelity, and switching discomfort differently [5]. Fourth, the structural finding that the shield-design space collapses into 11 behavioral classes is conditioned on *this* policy, ladder, and trace mix; alternative ladders (e.g., HDR with widely non-uniform VMAF gaps) could expand the discriminating power of the VMAF-aware fallback and warrant re-characterization. Fifth, the policy/shield co-design dependence we surface (Sec. 5.8) means that the recommended operating points are achievable only when the shipping client carries the shield; an operator who removes the shield silently is no longer at any of the points reported in Table 9. Finally, Genie is an oracle reference with perfect future trace knowledge, so its gap should be read as diagnostic context rather than a practically attainable target. Sixth, the synthetic 5G/mmWave outage suite isolates throughput shift but lacks stack realism; VMAF moreover ceases to be a sharp paired discriminator versus the legacy shield when sessions become overwhelmingly stall-limited, reinforcing that surrogate QoE and stall seconds remain the statistically reliable headline metrics there unless paired n is increased substantially or equivalence/non-inferiority tests are layered in [11].

Mitigation roadmap. Future work should evaluate the same Pareto-tunable shield family on live or packet-level emulation, add broader mobile/fixed-broadband trace families, report per-trace tail metrics in addition to means, and calibrate the catastrophic-ratio knob using deployment telemetry (e.g., per-CDN, per-region, per-time-of-day). Risk-sensitive objectives such as CVaR [35] and uncertainty-aware throughput prediction are natural extensions but should complement, not replace, explicit guard auditing. A particularly attractive direction is *adaptive* catastrophic-ratio scheduling: rather than a single fixed knob, an outer controller could move along the Pareto front in response to observed live QoE budgets.

7. Conclusion

We repositioned adaptive streaming as **Lagrangian CMDP surrogate training** complemented by a **VMAF-aware Pareto-tunable runtime shield**: a deterministic action projection that, when the catastrophic-download test fires, selects the highest-VMAF safe rung subject to a small VMAF-loss budget rather than uniformly decrementing the bitrate index. Under paired trace-driven evaluation on the primary FCC/Norway-style suite (Sec. 5), the proposed shield is the first ABR shield to our knowledge that achieves a *strict empirical three-way Pareto improvement* over the standard index-decrement projection on the same trained policy and identical trace seeds: paired QoE rises by +42.13 ($p=5.0 \times 10^{-3}$), paired rebuffering falls by $|-0.55|\%$ ($p=2.9 \times 10^{-3}$), and paired VMAF rises by +0.22 ($p=0.056$, marginal at $\alpha=0.05$ for $\tau=1.0$). A tighter VMAF-loss budget yields a strictly significant triple at $\alpha=0.05$: ++64.82 paired QoE, -0.69% rebuffering, and ++0.32 VMAF ($p=4.1 \times 10^{-2}$). Supplementary synthetic 5G/mmWave stress traces (Sec. 5.9) preserve significant paired QoE and rebuffer shifts versus legacy while paired VMAF is not cleanly resolved; we interpret this bundle as corroborative tail-risk stress testing rather than a second venue for VMAF-hypothesis refinement. A controlled sweep over 47 shield variants collapses into only 11 behavioral classes, identifying the VMAF-aware fallback (the projection *shape*) rather than the catastrophic-ratio threshold (the projection *level*) as the dominant Pareto driver. We further document a strong policy/shield co-design dependence: removing the shield at inference inflates rebuffering by $|-5.92|\%$ ($p=1.4 \times 10^{-10}$), arguing that constrained policies and their shields are operationally a single shippable unit. The contribution is therefore a deployable, auditable, and *empirically Pareto-improving*

runtime shield (strict nonparametric triple at $\tau=0.8$; recommended $\tau=1.0$ retains strong QoE/rebuffer gains) with two interpretable deployment knobs (VMAF-loss tolerance and catastrophic ratio)—rather than yet another single-point ABR controller. Future work targets differentiable risk metrics such as CVaR [35], online prediction uncertainty, adaptive scheduling of the VMAF-loss tolerance from live telemetry, and live/packet-level validation.

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Not applicable.

Data availability

The code, evaluation scripts, and exported CSV logs underpinning Tables 4 and 5 are provided as reproducibility artifacts in the repository referenced by this manuscript. Shield implementations are deterministic with parameters documented alongside the code to support third-party auditing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRedit authorship contribution statement

Saeed Zarbi: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Leili Farzinvash:** Supervision, Writing – review & editing. **Pedram Salehpour:** Conceptualization, Supervision, Project administration, Writing – review & editing.

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